# KEVIN O'DONNELL

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## EDUCATION

**Columbia University** New York, NY
***Master of Science in Biomedical Engineering**, Concentration in Robotics and Control* Dec 2024
Coursework: Biomed Innovation, Biostats, Robotics, Control, Mechatronics, Digital Mfg, Lab-To-Market

**University of California, Los Angeles (UCLA)** Los Angeles, CA
***Bachelor of Science in Honors Integrative Biology and Physiology*** Mar 2022
Awards: Department Honors, Alumni Scholarship National Finalist, Achievement Scholarship

## PROFESSIONAL EXPERIENCE

**ForUs (Generator Health)** Remote
Operations Specialist Feb 2026 - Present
- Automate workflows using Excel/Python to accelerate 100+ daily case resolutions and elevate quality.
- Train team on SOP, coordinate daily workloads, and collaborate with insurers, providers, and patients.

**Shape Memory Medical Inc.** San Jose, CA
Engineering Intern (cross-functional) Jun 2023 - Aug 2023
- Designed 3 patient-specific cardiac aneurysm models in CAD (SolidWorks), 3D-printed prototypes, and integrated them into a pulsatile flow system for embolization plug R&D Verification & Validation.
- Assisted in device delivery, system calibration, and sensor data collection during preclinical testing.
- Added manual activation control to manufacturing stent crimper circuitry, enhancing system reliability.
- Developed Minitab Design of Experiments to optimize laser welding, increasing tensile strength 367%

**Samuel Sia Laboratory - Microfluidics** New York, NY
Graduate Research Assistant Aug 2022 - Dec 2022
- Characterized microfluidic pump designs through prototype testing and experimental evaluation, streamlining sample processing and accelerating early COVID detection. Executed PCR testing.

**Colwell Laboratory of Circadian and Sleep Medicine** Los Angeles, CA
Honors Undergraduate Research Jul 2019 - Jun 2022
- Led a 3-year study/paper on adolescent circadian maturation in wild-type and BLOC-1-deficient mice.
- Developed experimental methods, revealing 5-day faster sleep-wake consolidation in females vs males
- Benchmarked wheel-running against infrared (IR) sensor performance through time-series data analysis of behavioral activity tracking and circadian dynamics characterization.

**Silk Road Medical Inc. (Acquired by Boston Scientific)** Minneapolis, MN
R&D Engineer Intern Jun 2021 - Sep 2021
- Engineered an Arduino carotid artery test bed with sensor components to assess pressure and flow for endovascular catheter/stent procedures, enabling real-time clinician feedback during evaluation.
- Developed post-market surveillance Python tool for root-cause analysis and new product development.

## PROJECTS (kevinod.com/Projects)

- Expandable endotracheal intubation model with Design Controls. Interviewed clinical staff and KOLs to translate user needs and risk analysis into design inputs and device requirements. Pitched the project with five-person team to physicians and investors; won best device design (CU Innovation 2022).
- Machine learning for healthcare datasets: malaria cell object detection, breast cancer treatment survival modeling, and cardiovascular risk prediction with fairness calibration (>95% AUC-ROC) (2025).
- Retinal needle commercialization plan, assessed technical and business viability with CU MBA (2023)
- Full-stack Python, API-based medication reminder for hospital EHR integration (UCLA YTP 2021).
- Robotic Arm with 5-DOF integrating ROS 2, MoveIt 2 path planning, Gazebo/RViz 2 simulation, motion planning, Raspberry Pi Bluetooth real-time control, and electromechanical integration (2025).

## SKILLS

**Engineering**: Design Controls, risk analysis, mechatronics, 3D printing, rapid prototyping, DOE
**Software**: CAD (SolidWorks), Minitab, Python, C++, machine learning (AI), statistics, data analysis